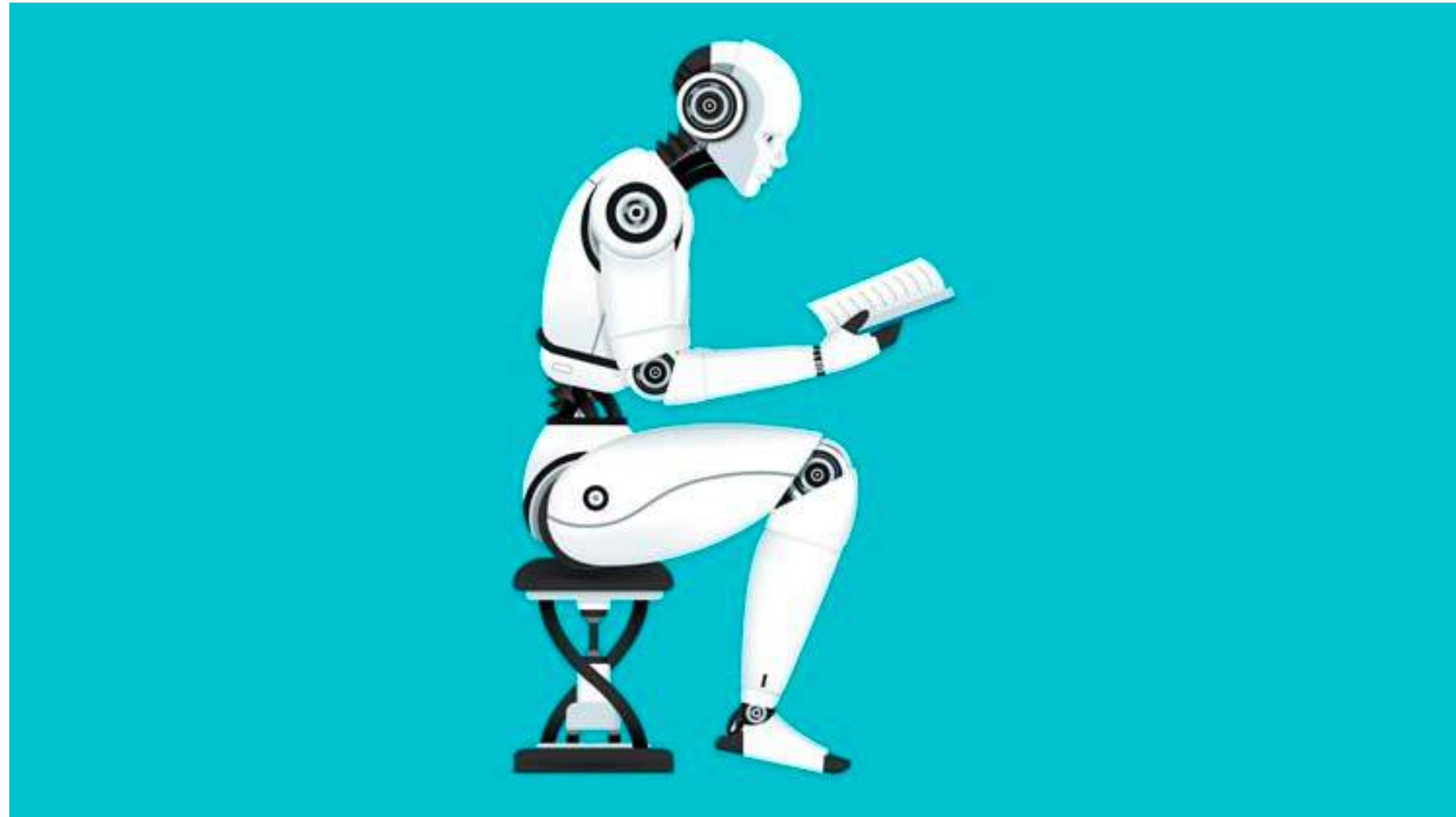




Machine Bias

Machine Learning | Enginyeria Informàtica

Santi Seguí | 2026-2027

What is a (ML) Classifier

Types of Classification Tasks

Binary Classification



Infected
Not infected

MultiClass Classification



Cat
Dog
Horse
Person
Fish
Bird
...

Multilabel Classification



Cat
Dog
Horse
Person
Fish
Bird
...

A ML classifier

- Learns from (historical) data.
- In many domains shows better accuracy than other approaches (also human decisions).
- However, these methods are **not-error free** (neither the humans are).

Learning from data

- The heavy dependence on data poses a new challenge though.
- The **data used for training** a machine learning algorithm are considered the **ground truth**.
- This means that during the learning phase, **these data constitute the comprehensive representation of the real world** which the algorithm seeks to approximate.

How we evaluate a Classifier

Example: We want to train a model that diagnoses Colon Cancer

Training:

Positive Cases: 10

Negative Cases: 90

Test:

Positive Cases: 10

Negative Cases: 90

Classification Error

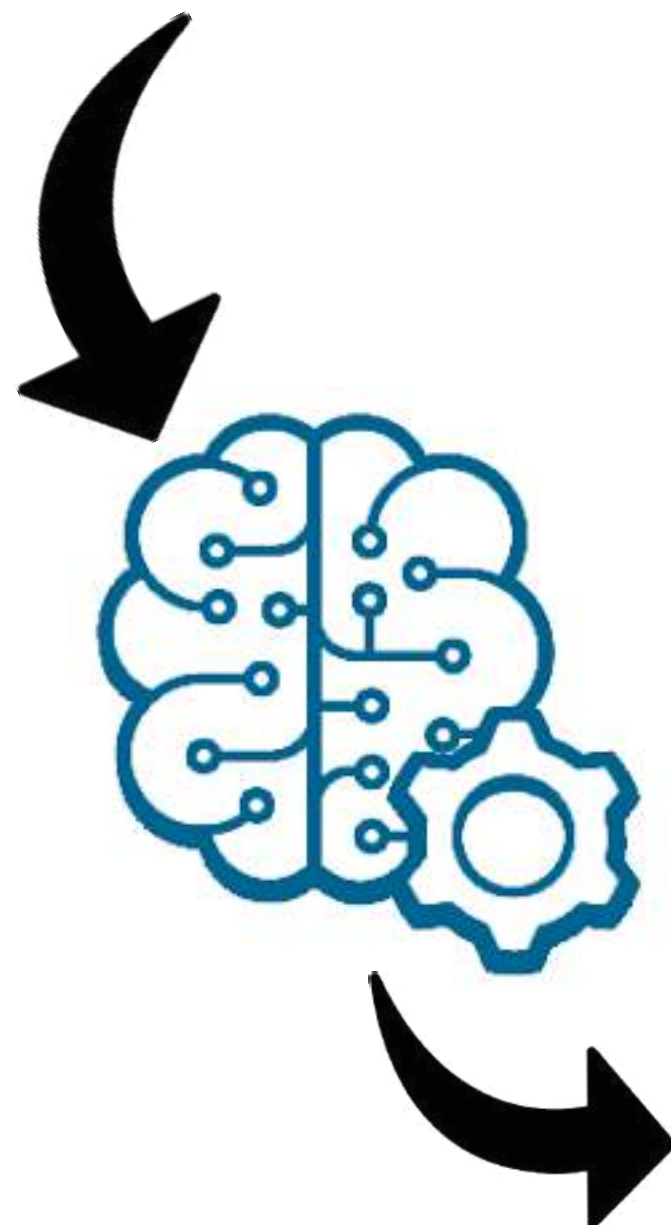
- **FP** = Negative considered Positive
- **FN** = Positive considered Negative
- **What is best, FP or FN?**
 - **It depends on the problem.**
 - A FP may lead a subject to undergo an unnecessary treatment
 - A FN may not receive an intervention when one would have been beneficial

Example: We want to train a model that diagnoses Colon Cancer

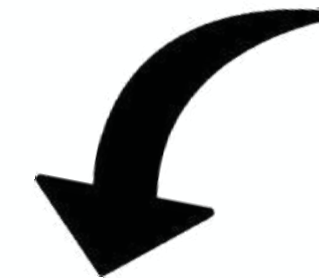
Training:
Positive Cases: 10
Negative Cases: 90

Test:
Positive Cases: 10
Negative Cases: 90

Train Model



Evaluate Model



— — — — — Result — — — — —

Positive Cases:

7 classified as Positive

3 classified as Negative

Negative Cases

81 classified as Negative

9 classified as Positive

Evaluation Measures

Accuracy or Classification Error

Confusion Matrix

Precision

Recall/Sensitivity

Specificity



$$\text{Accuracy} = \frac{TP + FP}{TP + FP + TN + FN}$$

$$(7 + 81) / 100 = 0.88$$

Evaluation Measures

Accuracy or Classification Error

Confusion Matrix

Precision

Recall/Sensitivity

Specificity



Real	Predicted	
	Positive	Negative
	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Real	Predicted	
	Positive	Negative
	Positive	Negative
Positive	7	3
Negative	9	81

Evaluation Measures

Accuracy or Classification Error

Confusion Matrix

Precision

Recall/Sensitivity

Specificity



$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\frac{7}{7 + 9} = \frac{7}{16} = 0.4375$$

Evaluation Measures

Accuracy or Classification Error

Confusion Matrix

Precision

Recall/Sensitivity

Specificity



$$\text{Recall} = \frac{TP}{TP + FN} = \frac{TP}{P}$$

$$\frac{7}{7 + 3} = \frac{7}{10} = 0.7$$

Evaluation Measures

Accuracy or Classification Error

Confusion Matrix

Precision

Recall/Sensitivity

Specificity

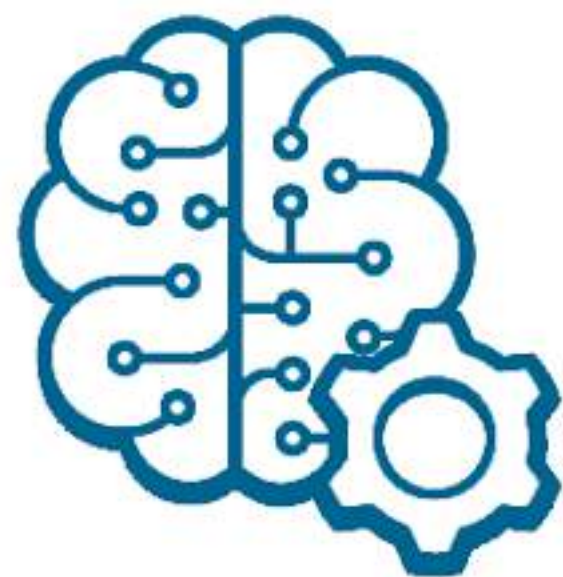


$$\textit{Specificity} = \frac{TN}{TN + FP} = \frac{TN}{N}$$

$$\frac{81}{81 + 9} = \frac{81}{90} = 0.9$$

Which is the best?

MODEL A



— — — — — Result — — — — —

Positive Cases:

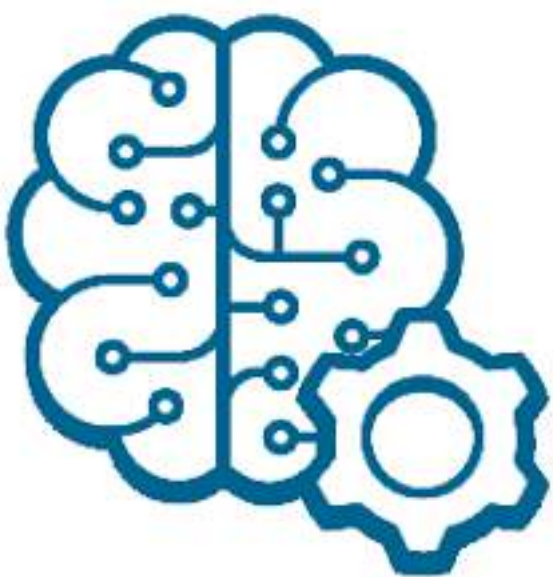
7 classified as Positive
3 classified as Negative

Negative Cases

81 classified as Negative
9 classified as Positive

	Predicted	
	Positive	Negative
Positive	7	3
Negative	9	81

MODEL B



— — — — — Result — — — — —

Positive Cases:

9 classified as Positive
1 classified as Negative

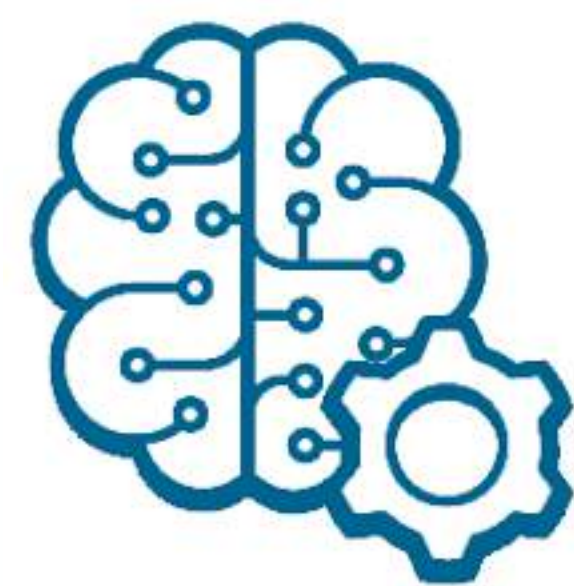
Negative Cases

78 classified as Negative
12 classified as Positive

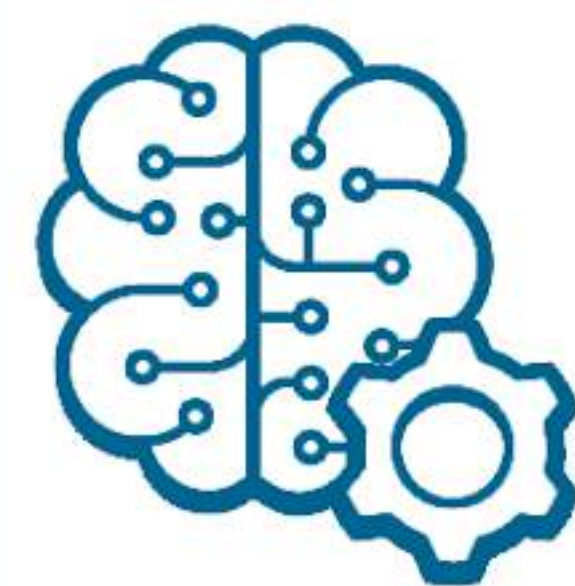
	Predicted	
	Positive	Negative
Positive	9	1
Negative	12	78

Which is the best?

MODEL A

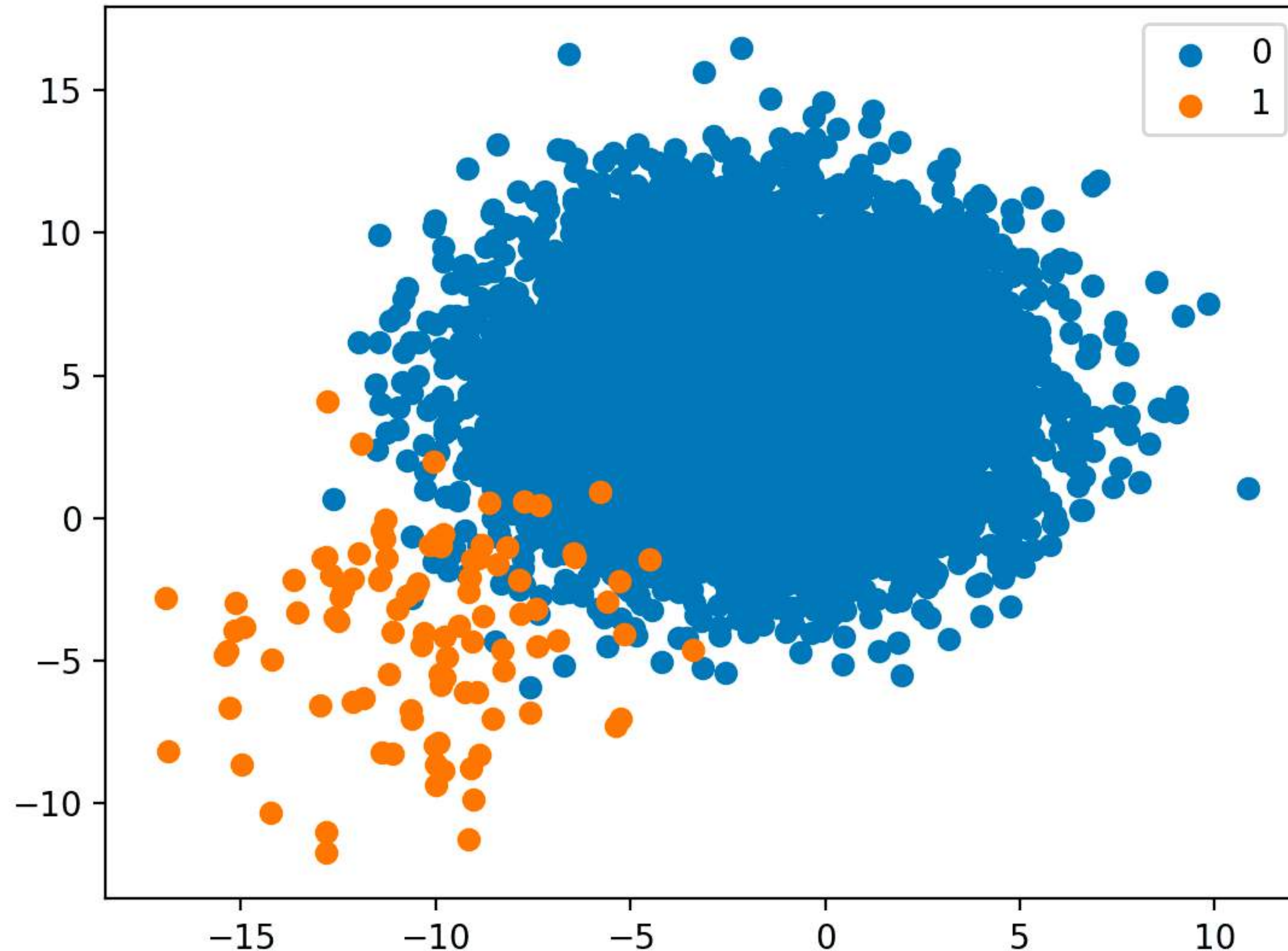


MODEL B

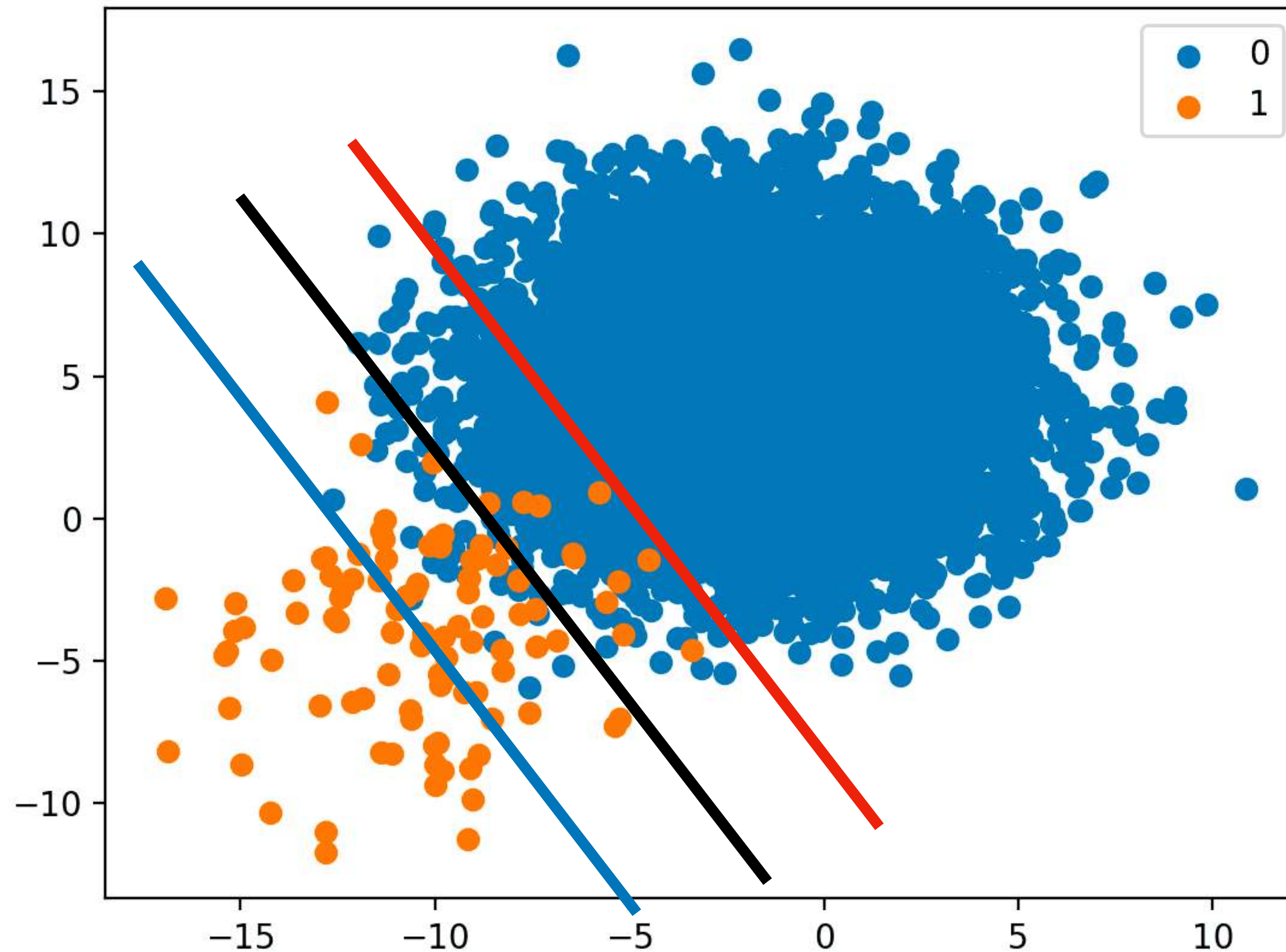


	Model A	Model B
Accuracy	0.88	0.87
Precision	0.43	0.42
Recall/Sensitivity	0.70	0.9
Specificity	0.90	0.86

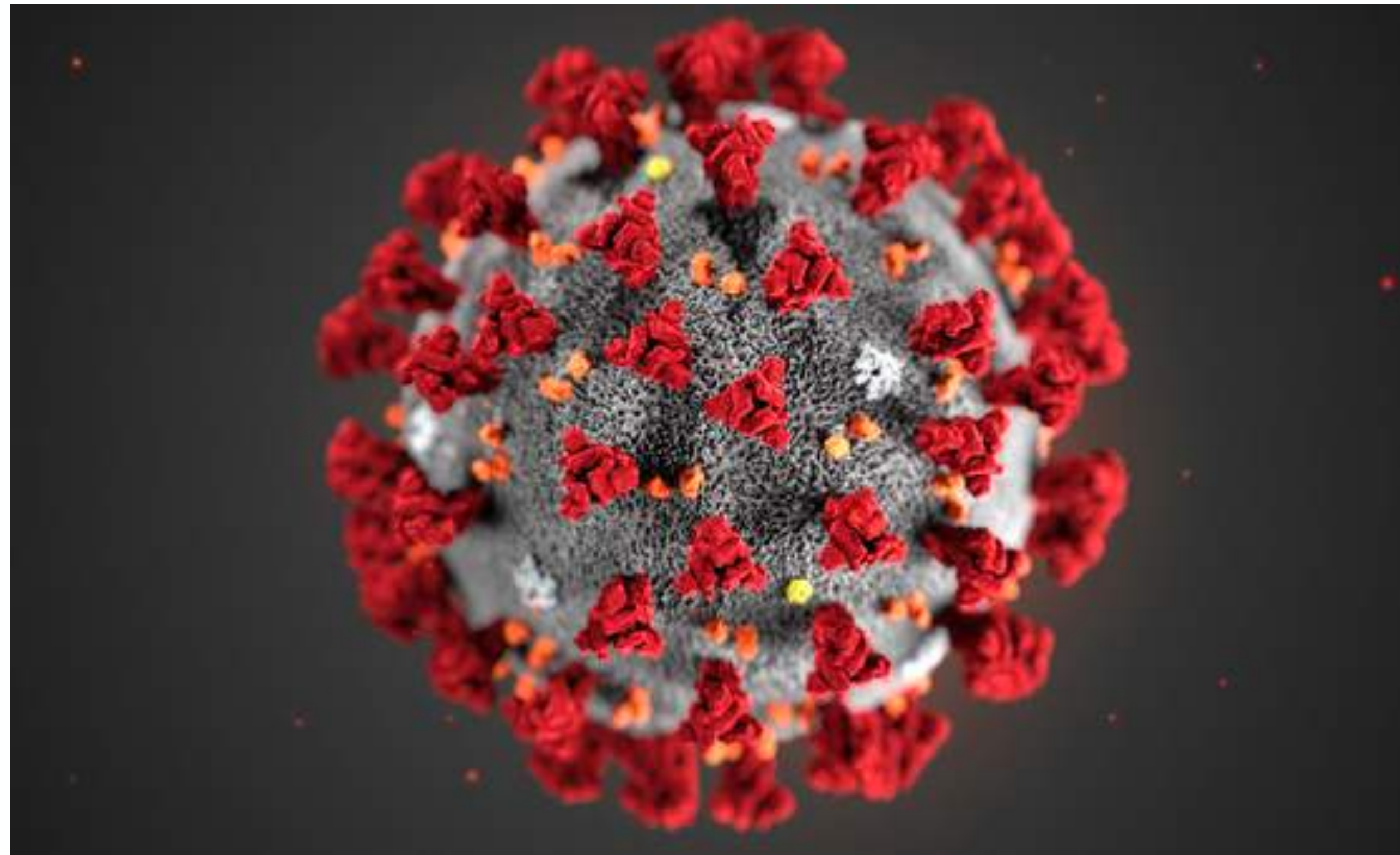
We can calibrate the model



I can calibrate the model



What is more important?



These models are used to **take decisions**

It is not just about accuracy...

Is our model fair?

This models are used to take decisions

- These decision can be critical and may affect our lives..
- The areas of application are extensive and include **medical diagnosis, university admission, loan allocation, recidivism prediction, recruitment**, online advertisement, face recognition, language translation, recommendation engines, fraud detection, credit limits, pricing and false news detection.

Learning from data

- The heavy dependence on data poses a new challenge though.
- The **data used for training** a machine learning algorithm are considered the **ground truth**.
- This means that during the learning phase, **these data constitute the comprehensive representation of the real world** which the algorithm seeks to approximate.
- If the **training data includes any kind of unwanted bias**, the **resulting algorithm** will **incorporate and enforce** it.
- Worse still, in the absence of robust explanations for the results, it is hardly possible for humans to recognise biased predictions of machine learning algorithms as such.

Ethical problems

- **Unwanted bias may happen to be directed against sensitive subgroups**, defined for instance by gender, ethnicity or age.
- People from one such group would be generally **disadvantaged** by the system.
- There is broad consensus in our society that making a distinction based on a personal characteristic which is usually not a matter of choice is unfair.
- Hence, anti-discrimination laws in plenty of legislations prohibit actions of this nature.
- Current legal standards rules to exclude any attribute which defines membership in a sensitive subgroup as feature from the data. For example, a user's gender may not be collected and processed in many scenarios. **This is not enough!**

Unwanted bias

- Two main sources for undesired bias have been identified in the machine learning pipeline:
 - **Training data are incorrect** or not sufficiently representative in certain aspects
 - **Training data may indeed faithfully** represent the **real world**, but the status quo **does not appear ideal**.
- Without correction, the machine learning algorithm would reproduce the current state and thus manifest an existing shortcoming.

²⁰ Translating from English to Turkish, then back to English injects gender stereotypes.**

The screenshot displays two sequential Google Translate sessions. In the first session, the source language is set to 'English' and the target to 'Turkish'. The input text is 'She is a doctor. He is a nurse.' The resulting Turkish translation is 'O bir doktor. O bir hemşire.' (O is a doctor. O is a nurse). In the second session, the source language is 'English' and the target is 'Turkish - detected'. The input text is the Turkish translation from the first session: 'O bir doktor. O bir hemşire'. The resulting English translation is 'He is a doctor. She is a nurse', which reintroduces the gender stereotypes from the original English text. The interface includes language selection buttons, a 'Translate' button, and character counts.

English Turkish Spanish Detect language ▾

English Turkish Spanish ▾ Translate

She is a doctor.
He is a nurse.

O bir doktor.
O bir hemşire.

English Turkish Spanish Turkish - detected ▾

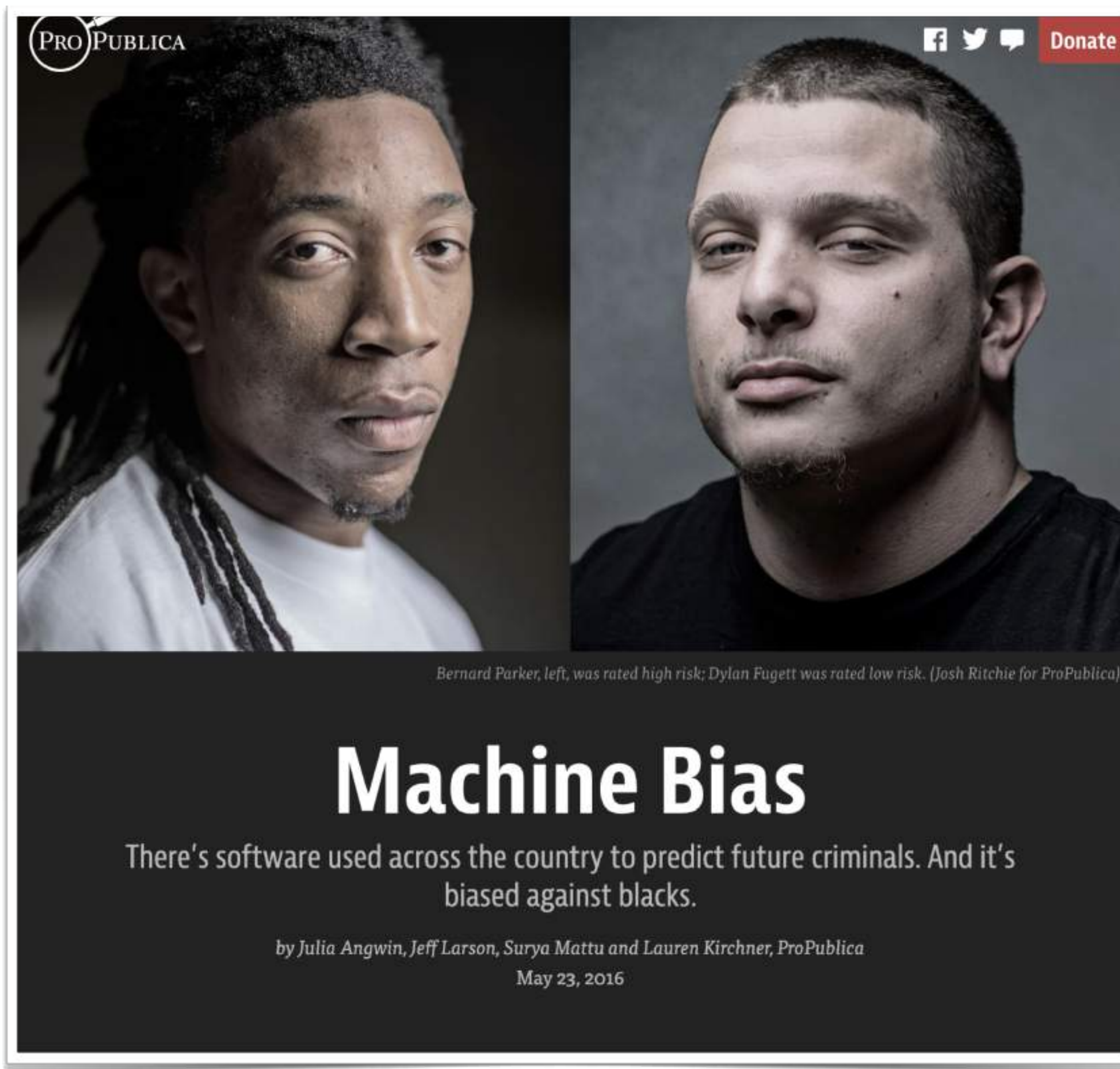
English Turkish Spanish ▾ Translate

O bir doktor.
O bir hemşire

He is a doctor.
She is a nurse ✓

CASE ANALYSIS: COMPAS





In May 2016, ProPublica, an investigative journalism organizations, published a piece called "**Machine Bias**", in which a predictive tool for predicting recidivism used in the US, **COMPAS**, was found to be **biased** against **African-Americans**

COMPAS CASE: A model to predict is **recidivism**

They looked at more than 10,000 criminal defendants in Broward County, Florida, and compared their predicted recidivism rates with the rate that actually occurred over a two-year period

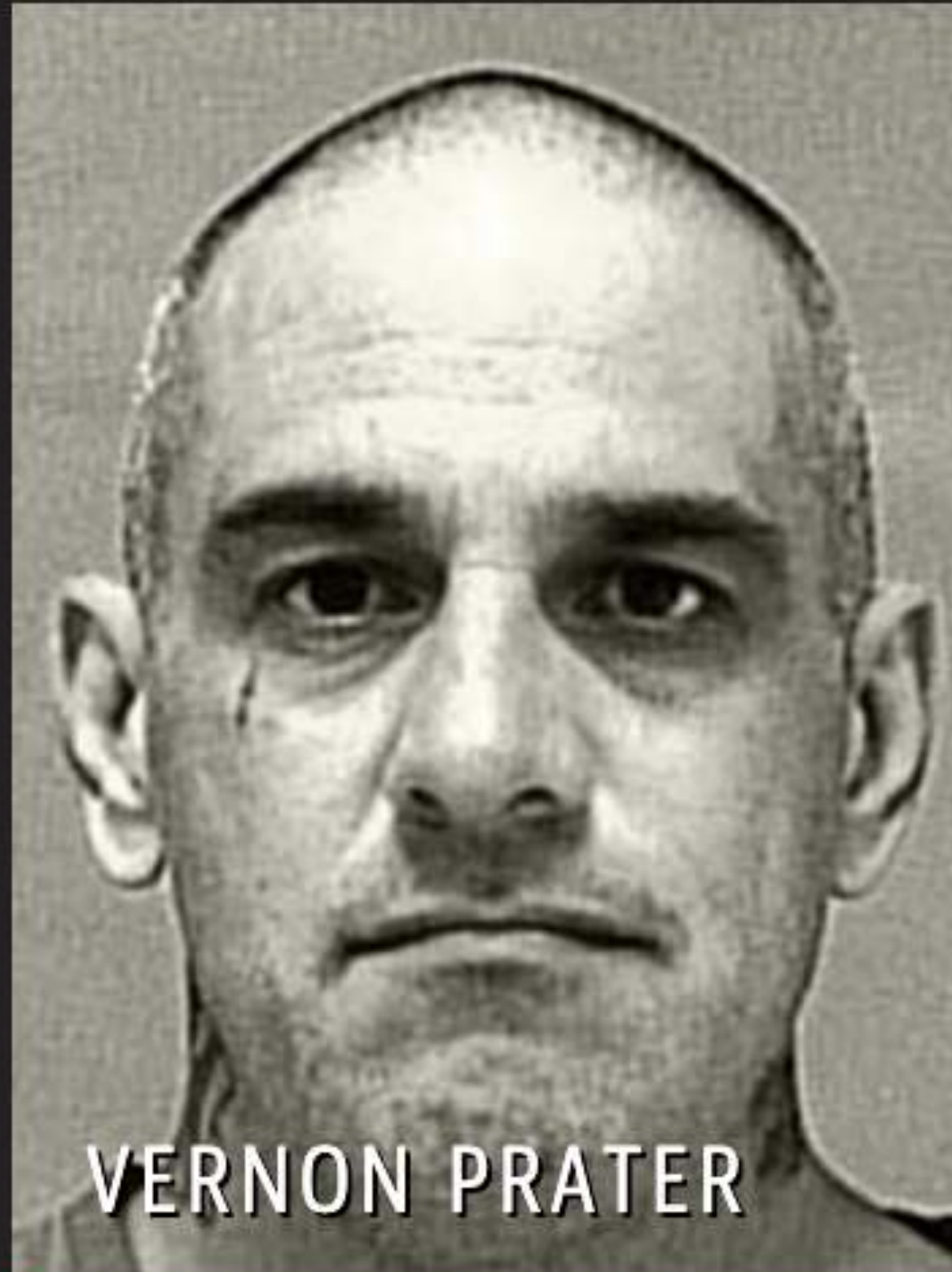
Real	Predicted	
	Low	High
	2681	1282
Not Re-offend	2681	1282
Re-offend	1216	2035

$$\text{Accuracy} = (2681 + 2035) / 7214 = 65.37$$



It has been demonstrated that this accuracy is higher than the one coming from judges

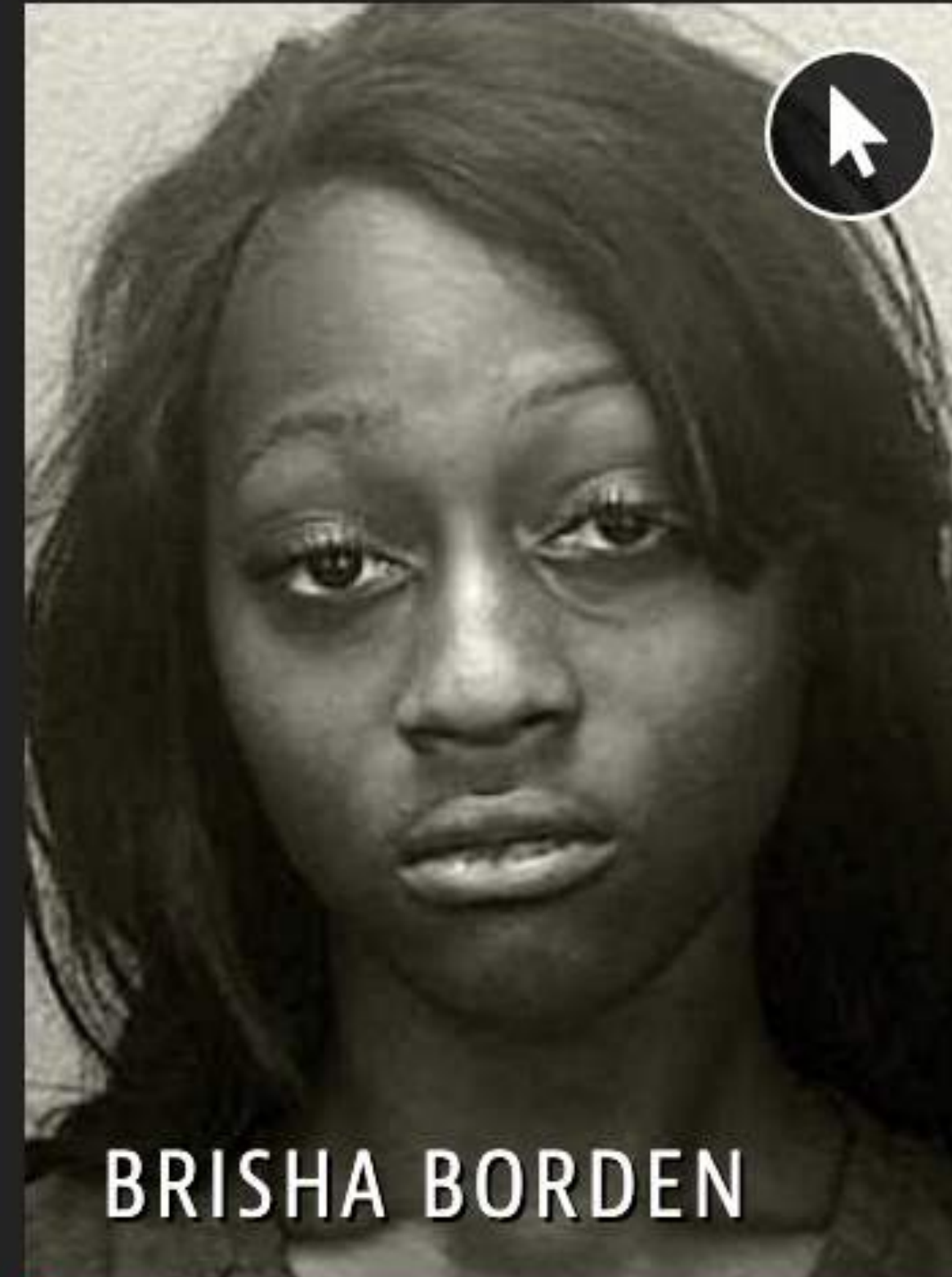
Two Petty Theft Arrests



VERNON PRATER

LOW RISK

3



BRISHA BORDEN

HIGH RISK

8

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

Two Petty Theft Arrests

VERNON PRATER

Prior Offenses

2 armed robberies, 1
attempted armed
robbery

Subsequent Offenses

1 grand theft

LOW RISK

3

BRISHA BORDEN

Prior Offenses

4 juvenile
misdemeanors

Subsequent Offenses

None

HIGH RISK

8

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

Two Petty Theft Arrests

VERNON PRATER

Prior Offenses

2 armed robberies, 1 attempted armed robbery

Subsequent Offenses

1 grand theft

LOW RISK

3

BRISHA BORDEN

Prior Offenses

4 juvenile misdemeanors

Subsequent Offenses

None

HIGH RISK

8

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

Prediction Fails Differently for Black Defendants

	WHITE	AFRICAN AMERICAN
Labeled Higher Risk, But Didn't Re-Offend	23.5%	44.9%
Labeled Lower Risk, Yet Did Re-Offend	47.7%	28.0%

Overall, Northpointe's assessment tool correctly predicts recidivism 61 percent of the time. But blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend. It makes the opposite mistake among whites: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes. (Source: ProPublica analysis of data from Broward County, Fla.)

Northpointe, the company that sells **COMPAS**, said in response that the test **was racially neutral**. To support that assertion, company officials pointed to another of our findings, which was that the rate of **accuracy** for COMPAS scores — about 60 percent — **was the same for black and white defendants**. The company said **it had devised the algorithm to achieve this goal**. A test that is correct in equal proportions for all groups cannot be biased, the company said.

The image is a screenshot of a ProPublica website article. At the top, the ProPublica logo is visible along with navigation links for 'ProPublica', 'Local Initiatives', and 'Data Store'. There are also social media icons for Facebook and Twitter, and a 'Donate' button. Below the logo, there are links for 'Graphics & Data', 'Newsletters', and 'About'. A search bar with the text 'Get the Big Story' and a 'Join' button is also present. The main navigation bar includes categories like 'Racial Justice', 'Technology', 'Criminal Justice', 'Regulation', and 'More...'. The article itself is titled 'MACHINE BIAS' and 'Bias in Criminal Risk Scores Is Mathematically Inevitable, Researchers Say'. The byline reads 'ProPublica's analysis of bias against black defendants in criminal risk scores has prompted research showing that the disparity can be addressed — if the algorithms focus on the fairness of outcomes.' The author is 'by Julia Angwin and Jeff Larson, Dec. 30, 2016, 4:44 p.m. EST'. The article features a large image of two men, one Black and one white, looking directly at the camera. To the right of the image is an advertisement for 'Pacsolver' with the text 'Resolvió PACs PECs. Aprueba sin Esfuerzo. Consigue soluciones rápidas y de calidad.' and a 'Pedir precios' button. Below the advertisement are social media links for Twitter, Facebook, YouTube, and RSS. At the bottom, there is a 'STAY INFORMED' section.

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MACHINE BIAS

Bias in Criminal Risk Scores Is Mathematically Inevitable, Researchers Say

ProPublica's analysis of bias against black defendants in criminal risk scores has prompted research showing that the disparity can be addressed — if the algorithms focus on the fairness of outcomes.

by Julia Angwin and Jeff Larson, Dec. 30, 2016, 4:44 p.m. EST

Ad

Resolvió PACs PECs. Aprueba sin Esfuerzo. Consigue soluciones rápidas y de calidad.

Pacsolver Pedir precios

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STAY INFORMED

**How can we evaluate if
there is any bias in our
model?**

Fairness

- There are many fairness approaches. But in general, they are related to one of the three major fairness criteria: **independence**, **separation**, and **sufficiency**.
- **Independence** requires prediction $\hat{Y}$ to be independent of A (group membership). If this requirement is met, it will fulfill demographic parity.
- **Separation** requires the prediction $\hat{Y}$ to be independent of the group membership (A) conditioning on the same outcome Y .
- **Sufficiency** requires Y to be independent of A conditioning on $\hat{Y}$.

Bias evaluation

- **Independence:** Sometimes called **demographic parity** or **statistical parity**

- $P(\hat{Y}|A=0) = P(\hat{Y}|A=1)$

- **Separation (Recall is more important):**

- **False Negative** is **more sensitive** (Sometimes called **equalised opportunities**)

- $P(\hat{Y}=1|Y=1,A=0) = P(\hat{Y}=1|Y=1,A=1)$

- **False Positive** is **more sensitive** (Sometimes called **predictive equality**)

- $P(\hat{Y}=1|Y=0,A=0) = P(\hat{Y}=1|Y=0,A=1)$

- **False Positive + False Negative** (Sometimes called **equalized odds**)

- $P(\hat{Y}=y|Y=y,A=0) = P(\hat{Y}=y|Y=y,A=1) \forall y \in \{0,1\}$

- **Sufficiency (Precision is more important):**

- **False Positive** is **more sensitive** (Sometimes called **predictive parity**)

- $P(Y=1|\hat{Y}=1,A=0) = P(Y=1|\hat{Y}=1,A=1)$

Demographic parity

20 applicants (50% from **Group A**)
14 approvals (50% from **Group A**)

		PREDICTED	
		Deny	Approve
TRUE	GROUP A	1	2
	GROUP B	2	5

		PREDICTED	
		Deny	Approve
TRUE	GROUP A	2	4
	GROUP B	1	3

Equal opportunity

Group A: 66% true positive rate: $4/(4+2)$
Group B: 66% true positive rate: $2/(1+2)$

		PREDICTED	
		Deny	Approve
TRUE	GROUP A	3	1
	GROUP B	2	4

		PREDICTED	
		Deny	Approve
TRUE	GROUP A	6	1
	GROUP B	1	2

Equal accuracy

Group A: 80% accurate: $(6+2)/10$
Group B: 80% accurate: $(4+4)/10$

		PREDICTED	
		Deny	Approve
TRUE	GROUP A	2	1
	GROUP B	1	6

		PREDICTED	
		Deny	Approve
TRUE	GROUP A	4	2
	GROUP B	0	4

How should we evaluate if **COMPAS** model is **fair**?

		PREDICTED	
		Low	High
TRUE	No	1139	349
	Yes	461	505

		PREDICTED	
		Low	High
TRUE	No	990	805
	Yes	532	1369

Bias evaluation

- **Independence:** Sometimes called **demographic parity** or **statistical parity**

- $P(\hat{Y}|A = 0) = P(\hat{Y}|A = 1)$

- **Separation (Recall is more important):**

- **False Negative** is **more sensitive** (Sometimes called **equalised opportunities**)

- $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$

- **False Positive** is **more sensitive** (Sometimes called **predictive equality**)

- $P(\hat{Y} = 1 | Y = 0, A = 0) = P(\hat{Y} = 1 | Y = 0, A = 1)$

- **False Positive + False Negative** (Sometimes called **equalized odds**)

- $P(\hat{Y} = y | Y = y, A = 0) = P(\hat{Y} = y | Y = y, A = 1) \forall y \in \{0,1\}$

- **Sufficiency (Precision is more important):**

- **False Positive** is **more sensitive** (Sometimes called **predictive parity**)

- $P(Y = 1 | \hat{Y} = 1, A = 0) = P(Y = 1 | \hat{Y} = 1, A = 1)$

Demographic parity

Group White: $P(\text{High} | \text{White}) = 35\%$
Group Black: $P(\text{High} | \text{Black}) = 59\%$

		PREDICTED	
		Low	High
TRUE	No	1139	349
	Yes	461	505

		PREDICTED	
		Low	High
TRUE	No	990	805
	Yes	532	1369

Equalised opportunities

Group White: 52%
Group Black: 72%

		PREDICTED	
		Low	High
TRUE	No	1139	349
	Yes	461	505

		PREDICTED	
		Low	High
TRUE	No	990	805
	Yes	532	1369

Predictive equality

Group White: 23%
Group Black: 45%

		PREDICTED	
		Low	High
TRUE	No	1139	349
	Yes	461	505

		PREDICTED	
		Low	High
TRUE	No	990	805
	Yes	532	1369

Equalized odds

Group White: 64%
Group Black: 63%

		PREDICTED	
		Low	High
TRUE	No	1139	349
	Yes	461	505

		PREDICTED	
		Low	High
TRUE	No	990	805
	Yes	532	1369

Predictive parity

Group White: 59%
Group Black: 63%

If you are interested to learn more:

Towards the Right Kind of Fairness in AI

A guide on the different metrics. And the tool “Fairness Compass”
to choose the best option for your project.

Boris Ruf and Marcin Detyniecki*

AXA GO, AI Research, Paris, France

30 Sep 2021

Should we allow a model like this?

Yes or **Not**? why?

There is an epidemic.

A new treatment has been discovered.

If you are infected and you get the treatment you heal!

If you get this treatment and you are not infected you can die.

We have a model that predicts if you are infected. This model is not 100% accurate... :(
but will save lots of lives

What do we prefer?
an **Accurate** or a Fair Model?

Accurate model

Saves 85% lives

(95% women & 75% men)

vs.

Fair model

Saves 80% lives

(80% women & 80% men)